罗马混凝土的千年之谜

The Secret of Rome's Undying Concrete



古罗马混凝土为何如此耐久？

英文原文

Nearly 1,900 years after its **completion**, in about AD 125–128, the Pantheon's **dome** still **crowns** the **vast chamber** below. About 43 metres **in diameter**, it remains the world's largest **unreinforced concrete** dome. It is an **astonishing feat** of **architecture**, and a **stubborn puzzle**.

Modern concrete is usually **reinforced with embedded steel bars**, which are **prone to corrosion**. As the steel **rusts**, it **swells** and **fractures** the concrete around it, and many **structures** are designed for a **lifespan** of around 50 years. Roman concrete is not stronger than its modern **counterpart**; its **merit** lies in **durability**.

Yet the ancient material seemed to have a **shortcoming**. It is **dotted with** white **chunks** of lime, known as lime **clasts**. Experts long **ascribed** these **lumps to sloppy** mixing or **inferior raw ingredients**, a **defect** that, they **assumed**, a more **conscientious** builder would have avoided.

A study published in Science Advances in January 2023 offered a different **interpretation**. The clasts, it suggested, were a **by-product** of "hot mixing" with **quicklime**, a **reactive** lime that **heats up** in water. When water **seeps into** a **crack**, the clasts **dissolve** and their **calcium** recrystallises as calcium carbonate, **plugging** the **gap**. In tests, water **ceased to trickle** through cracked hot-mixed samples **within weeks**, but kept flowing through samples without quicklime.

The **implications extend beyond archaeology**. Cement **manufacture accounts for** about 8 per cent of global carbon dioxide **emissions**, the **think-tank** Chatham House reported in June 2018. Concrete that **mends** its cracks could last longer, making **demolition** and rebuilding rarer and **trimming** emissions. The white lumps once **scorned** as signs of **clumsy** workmanship have **prompted** a **rethink** of construction.

中文译文

万神殿约于公元125至128年竣工，在其竣工近1,900年后，它的穹顶依然位于下方宏大的殿堂的顶部。穹顶直径约43米，至今仍是世界上最大的无钢筋混凝土穹顶。它是建筑史上令人惊叹的壮举，也是一道难以解开的谜题。

现代混凝土通常靠嵌入其中的钢筋加固，而钢筋容易发生锈蚀。随着钢筋生锈，它会膨胀，并使周围的混凝土断裂；而且许多建筑被设计为约50年的使用寿命。罗马混凝土并不比现代的同等对应物更坚固；它的长处在于耐久性。

然而，这种古代材料似乎有个缺点。它的内部散布着白色的石灰小块，称为石灰碎屑。专家长期以来把这些块状物归因于马马虎虎的搅拌或者是比较次的原材料。这种缺陷，他们认为，换一个更加认真负责的建筑工就能避免。

2023年1月发表于《科学进展》（Science Advances）的一项研究提出了不同的解释。这些碎屑，该研究认为，是与生石灰“热混合”的副产品：一种遇水会变热的活性强的石灰。当水渗入裂缝时，碎屑会溶解，它们的钙会重新结晶为碳酸钙，从而堵住缝隙。在试验中，水在几周之内便不再细细流过开裂的热混合样本，但却持续流过不含生石灰的样本。

这项发现的意义超出了考古学的范围。水泥生产占据全球二氧化碳排放量的约8%，智囊团查塔姆研究所于2018年6月报告说。能自行修补裂缝的混凝土可能更经久耐用，从而使拆除与重建减少，并削减排放。这些白色块状物曾被轻视为拙劣的手艺的标志，如今却促使人们重新思考建造之道。

词汇与表达

completion n.
竣工

dome n.
穹顶

crowns v.
位于……的顶部

vast adj.
宏大的

chamber n.
殿堂

in diameter n.
直径

unreinforced concrete n.
无钢筋混凝土

astonishing adj.
令人惊叹的

feat n.
壮举
It is an astonishing feat of

architecture n.
建筑

stubborn adj.
难以解开的

puzzle n.
谜题

reinforced with v.
靠……加固

embedded adj.
嵌入

steel bars n.
钢筋

prone to adj.
容易
which are prone to corrosion.

corrosion n.
锈蚀
which are prone to corrosion.

rusts v.
生锈

swells v.
膨胀

fractures v.
使……断裂

structures n.
建筑

lifespan n.
使用寿命
for a lifespan of around 50 years.

词汇与表达（续）

counterpart n.
同等对应物
stronger than its modern counterpart;

merit n.
长处

durability n.
耐久性
its merit lies in durability.

shortcoming n.
缺点
material seemed to have a shortcoming.

dotted with adj.
散布

chunks n.
小块

clasts n.
碎屑

ascribed to v.
把……归因于

lumps n.
块状物

sloppy adj.
马马虎虎的

inferior adj.
比较次的
mixing or inferior raw ingredients,

raw ingredients n.
原材料

defect n.
缺陷

assumed v.
认为

conscientious adj.
认真负责的
more conscientious builder would have

interpretation n.
解释
offered a different interpretation.

by-product n.
副产品
were a by-product of "hot mixing" with

quicklime n.
生石灰
of "hot mixing" with quicklime,

reactive adj.
活性强的
reactive lime that heats up in water.

heats up v.
变热

seeps into v.
渗入
When water seeps into a crack,

词汇与表达（续）

crack n.
裂缝

dissolve v.
溶解
the clasts dissolve and their calcium

calcium n.
钙

plugging v.
堵住

gap n.
缝隙

ceased to v.
不再

trickle v.
细细流过

within weeks prep.
在几周之内

implications n.
意义
The implications extend beyond

extend beyond v.
超出了……的范围
The implications extend beyond

archaeology n.
考古学

manufacture n.
生产

accounts for v.
占据
Cement manufacture accounts for about

emissions n.
排放
of global carbon dioxide emissions,

think-tank n.
智囊团

mends v.
修补

demolition n.
拆除
making demolition and rebuilding rarer

trimming v.
削减

scorned v.
轻视

clumsy adj.
拙劣的

prompted v.
促使
have prompted a rethink of

rethink n.
重新思考

文章简述

- 耐久：万神殿直径约43米、没用钢筋加固的穹顶已存近1900年。罗马混凝土的优势是能用得久，并非强度高于现代混凝土。
- 疑点：现代钢筋生锈后会膨胀，使周围的混凝土断裂；罗马材料中的白色石灰块，过去却被当成搅拌差或原料劣的缺陷。
- 新解：2023年研究认为，使用生石灰的热混合过程留下这些石灰块；水渗入裂缝后，石灰块溶解，其中的钙重新形成碳酸钙结晶，堵住裂缝。
- 试验：开裂的热混合样本几周内不再渗水，未加生石灰的对照仍漏水，支持材料能自行补缝的解释。
- 前景：水泥生产占据全球二氧化碳排放量的约8%；若能延长建筑寿命，就有望减少拆除、重建及相关排放。